How a Database Connects to a Website

From one click in the browser to rows in a table and back again, in five steps.

THE JOURNEY OF ONE REQUEST

1

USER / BROWSER

User submits a form or opens a page in the browser

GET · POST

2

WEB SERVER

Receives the HTTP request and routes it onward

Nginx · Apache

3

BACKEND / APPLICATION

Validates input and runs the application logic

Node.js · PHP · Python

never trust raw user input

4

DATABASE

Runs the SQL query and returns the matching data

MySQL · PostgreSQL · MongoDB

this is where data is fetched

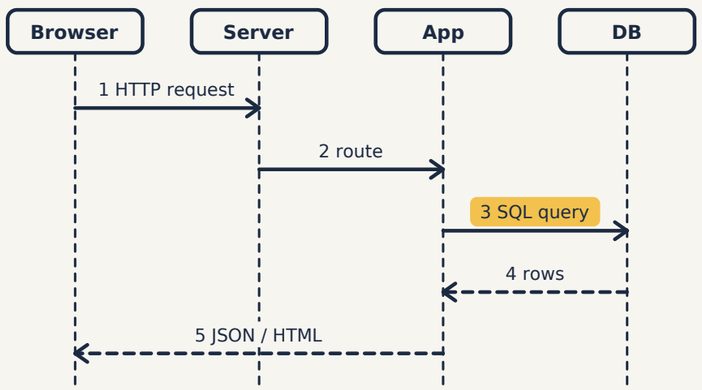
5

RESPONSE

Data becomes JSON or HTML and is shown on the page

JSON · HTML

Request-Response Cycle



Example Query

```
SELECT * FROM users
WHERE id = 1;
```

result set

id	name	email
1	Asha	asha@example.com

1 row returned

one user, not the whole table

Security & Performance

Prepared statements
Bind values, never concatenate SQL

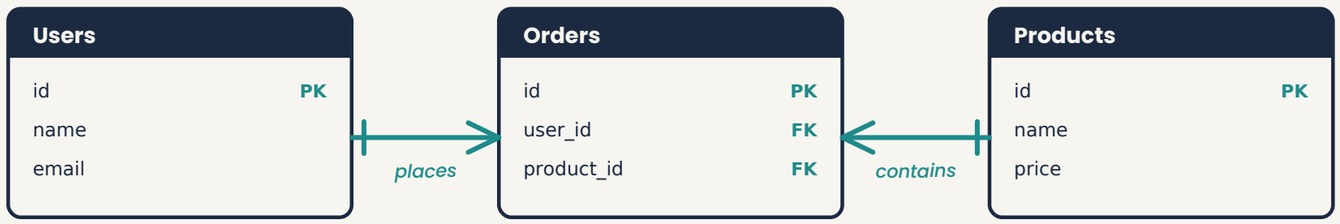
Authentication
Hash passwords, verify sessions or tokens

Connection pooling
Reuse open connections for faster requests

Indexing
Speed up lookups on id and email

Backups
Scheduled copies, restores tested

Data model: Users – Orders – Products



Real-world example: login on a website

The server runs a prepared query on the users table, compares the password hash, then starts a session so the next page knows you.

Login form → **SELECT by email** → **Hash check** → **Session**

same pattern powers sign-ups, carts and search